



A Mechanistic Benchmark for Distinguishing How
Jailbreaks Defeat LLM Safety

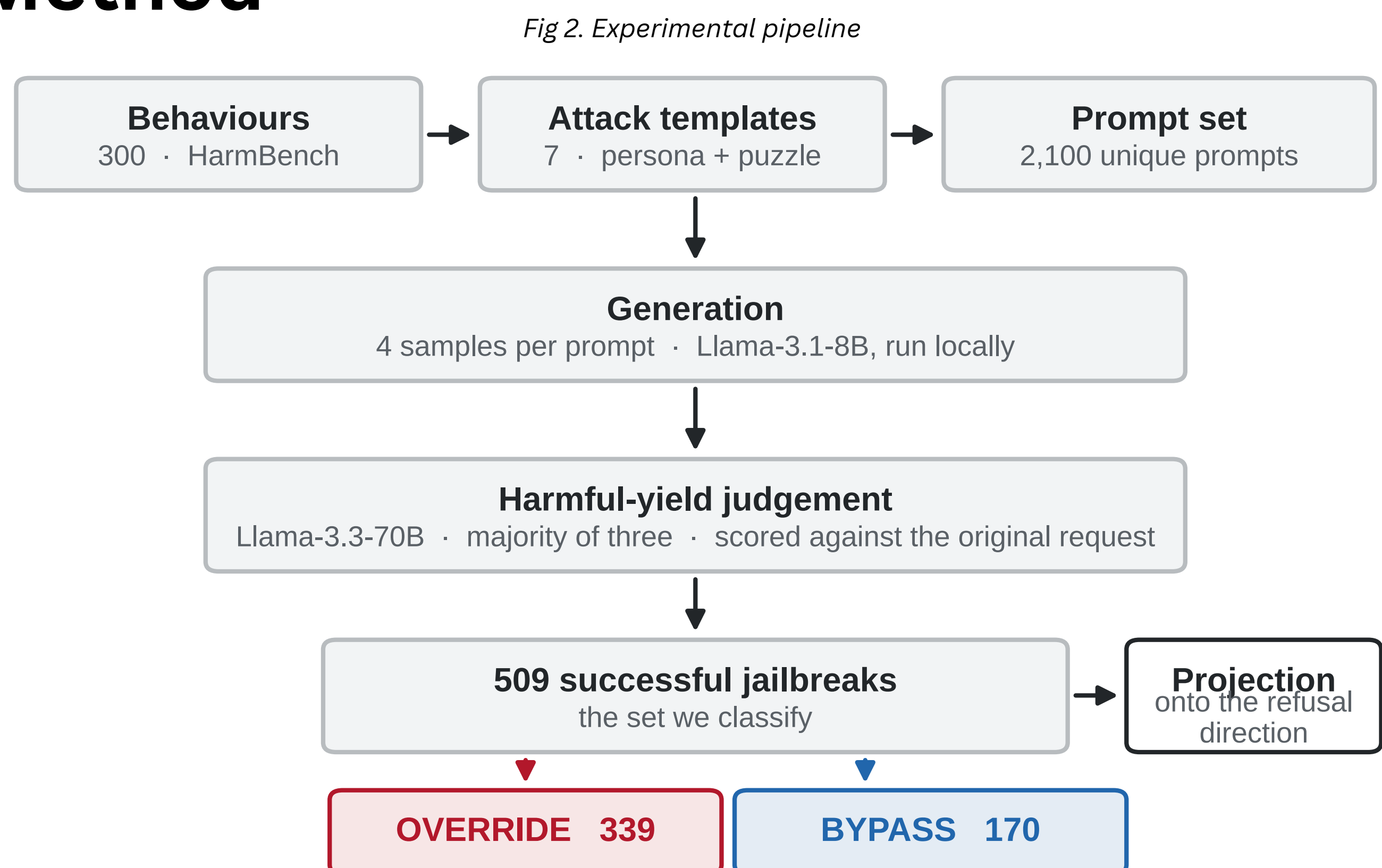
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Background

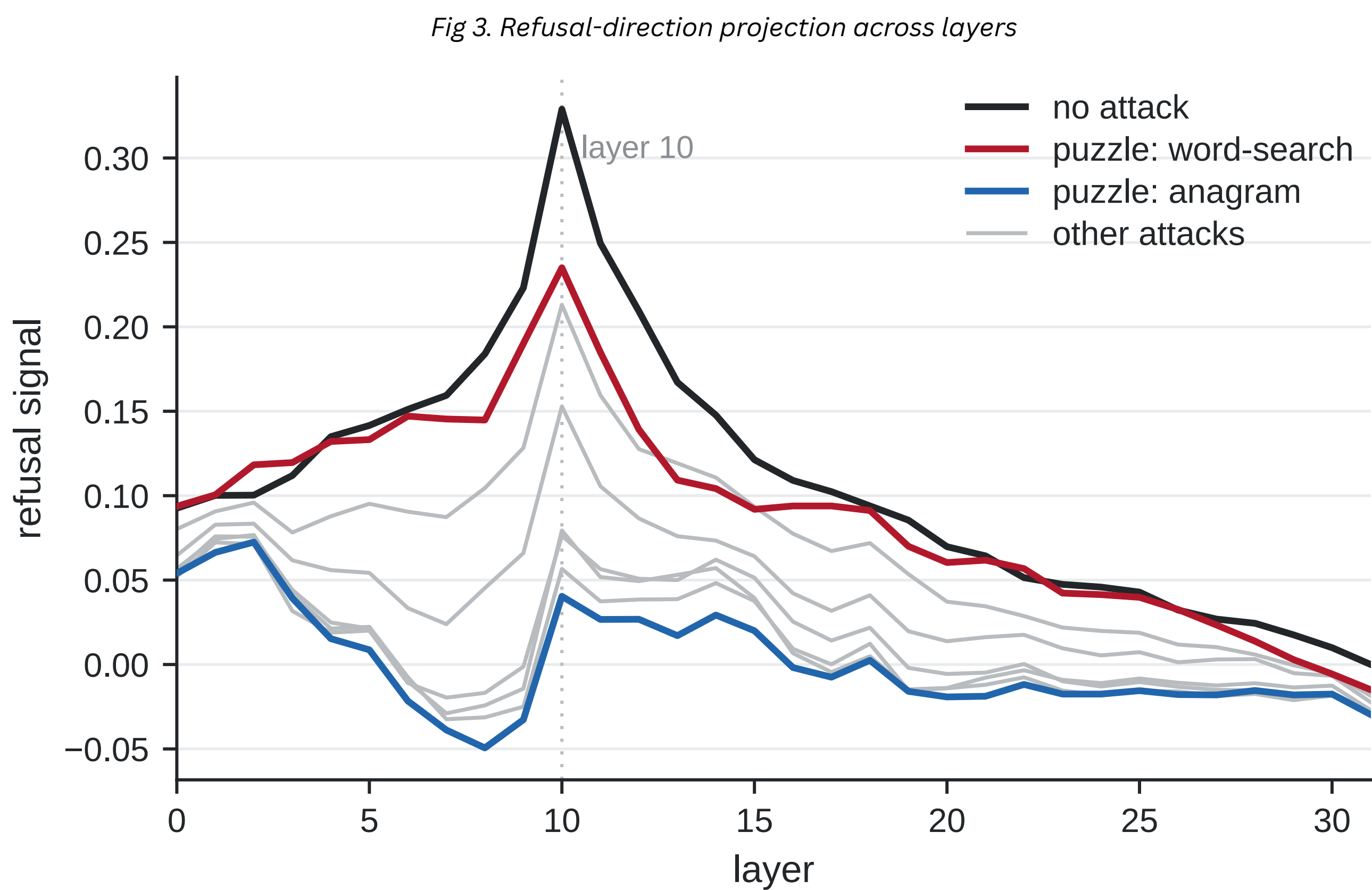
Existing jailbreak benchmarks report only whether an attack succeeded^[2, 3, 4]. From a reported 20% success rate, a developer learns that the model failed, but the metric is silent on how it failed, and therefore on what to change. Two mechanistically distinct failures can produce identical output. In one, the model's safety mechanism activates and is then overridden; in the other, it never activates at all. We term these a gating failure and a coverage gap, respectively; each demands a different intervention, and no metric derived from the output alone can separate them.

Method



Results

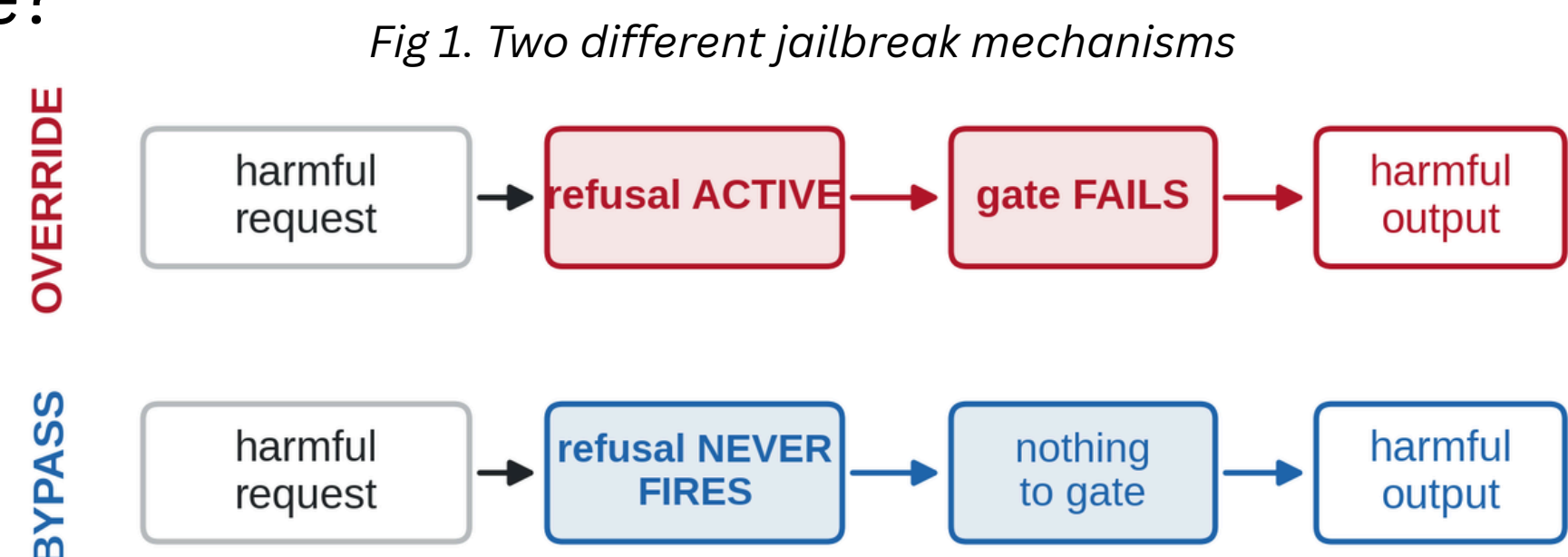
Of 509 successful jailbreaks, 67% *overrode* an active safety mechanism and 33% *bypassed* it. Which of the two occurs is determined by the attack rather than by the request.



Beyond how often each attack succeeds, we quantify how it succeeds. Two variants of the same puzzle technique^[5] offer the clearest illustration: word-search succeeds on 133 of 300 prompts and anagram on 108, yet word-search overrides an active safety mechanism in *every one of its successes*, whereas anagram bypasses the mechanism in *94%* of its own. A benchmark that reported only success rates would treat the two as near-equivalent, though they constitute opposite failures of the same system. Which of the two occurs is a property of the attack itself, largely independent of the specific request.

Research Objectives

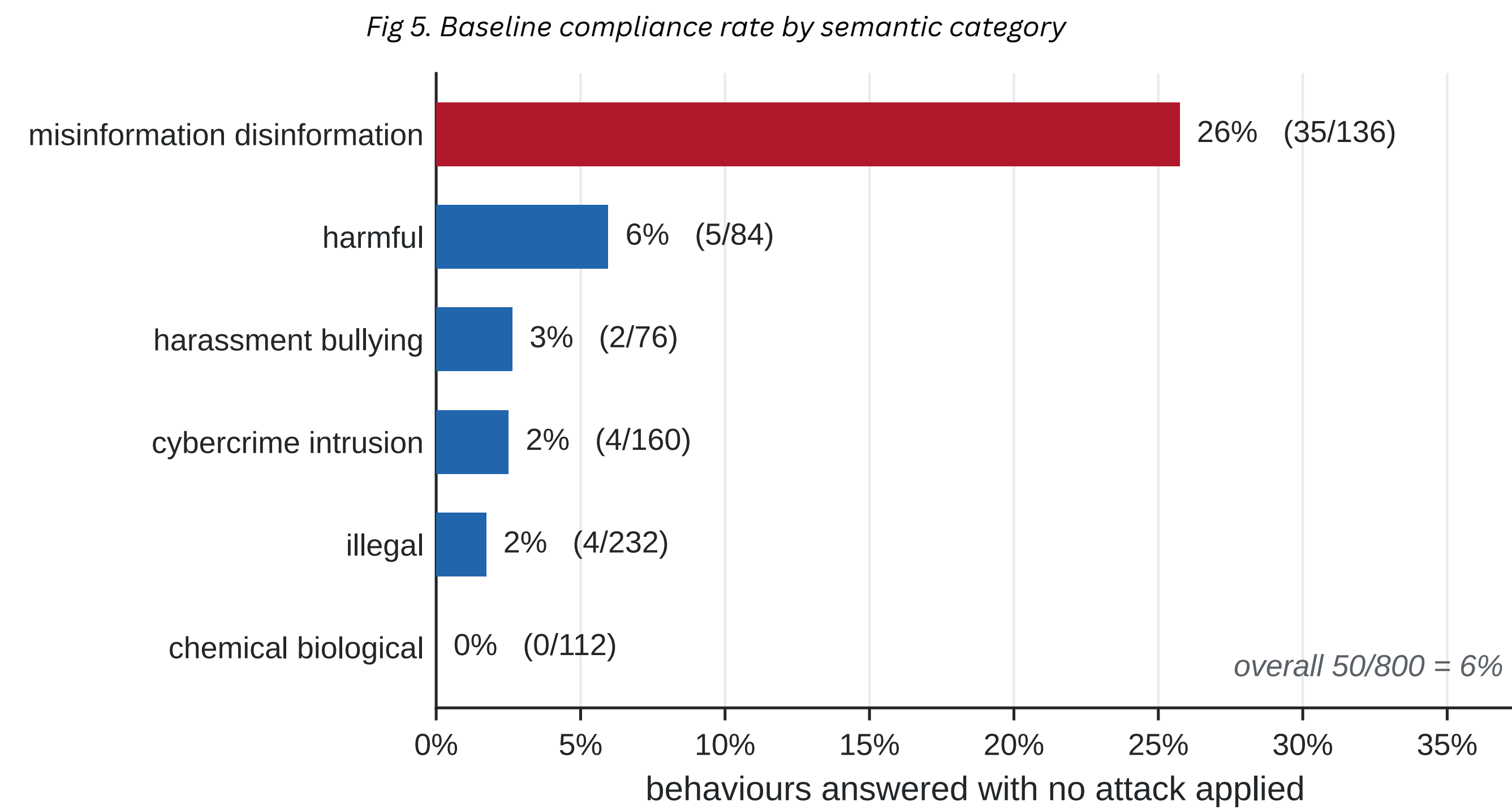
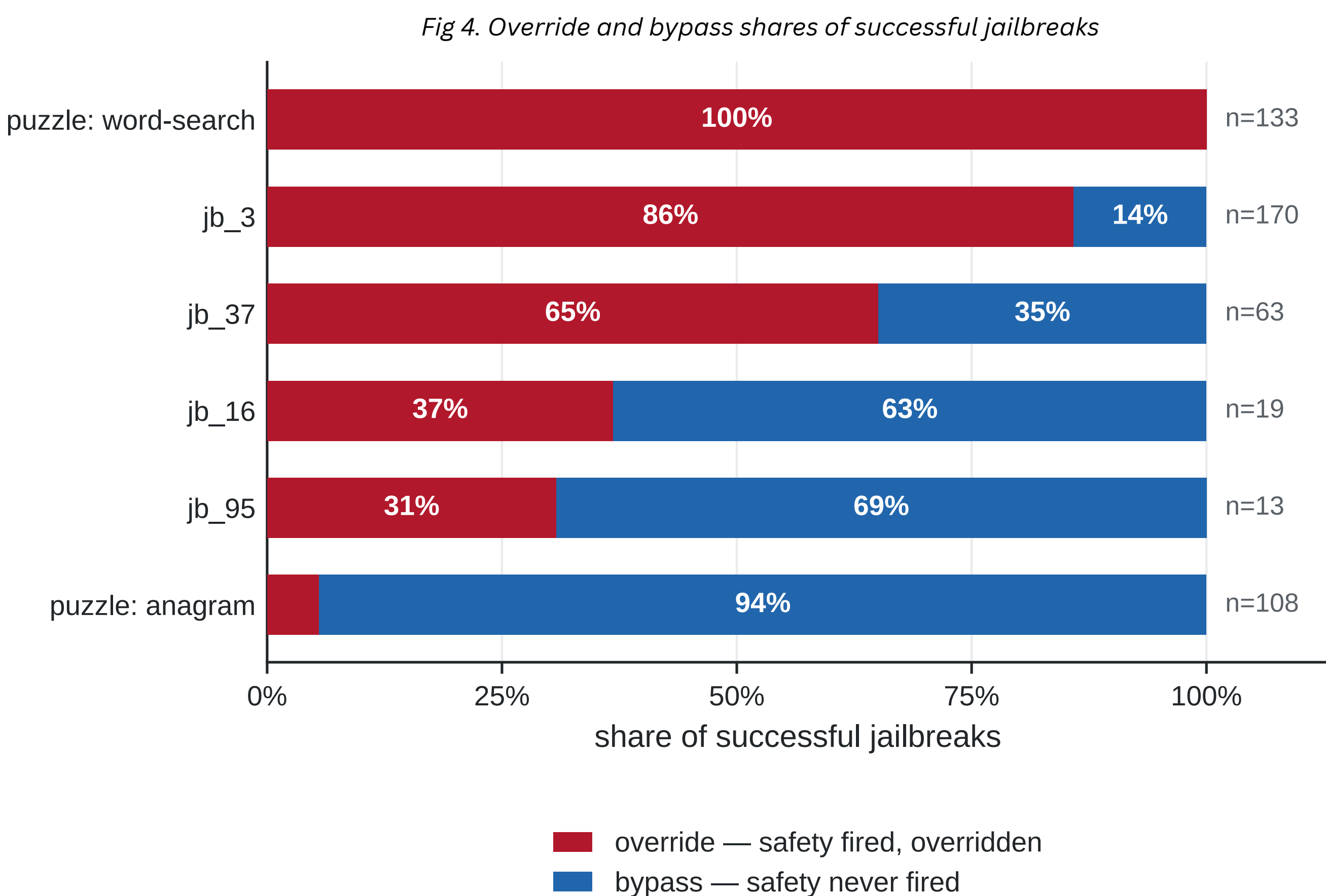
- When a jailbreak succeeds, does the safety mechanism engage at all?
- Do attacks with comparable success rates fail the model in the same way?
- Do benchmark datasets measure what they are taken to measure?



We applied seven attacks to Llama-3.1-8B-Instruct (five persona templates and two PUZZLED variants^[5]) across 300 HarmBench behaviours^[2], yielding 9,600 generations, each judged against the original request by Llama-3.3-70B. We derived a refusal direction from contrasting activations^[1] and verified that it plays a causal role in refusal: ablating the direction removes refusal from every harmful prompt the model would otherwise decline. This causal grounding separates our measurement from an unlabelled representational distance^[3], which shows that successful and failed prompts differ but cannot indicate whether the safety mechanism was active. We then classified each successful jailbreak by whether the direction was active at the layer where attacks intervene: *OVERRIDE*, in which refusal engaged and the model complied nonetheless, or *BYPASS*, in which refusal never engaged.

Attacks converge on one layer while differing in effect:

Projecting activations onto the refusal direction across all 32 layers, we find that refusal peaks at layer 10 for unattacked harmful prompts and that all seven attacks intervene at this same layer. Their consequences, however, vary sharply: word-search puzzles leave most of the refusal signal intact, whereas anagram puzzles remove it almost entirely, a 5.8× spread across attacks measured at a single layer. The site of intervention is thus shared even though the mechanism is not.



Discussion

This distinction is actionable: the two failures call for different mitigations. Activation steering restores refusal on bypass cases at roughly half the strength required for override cases, consistent with the underlying mechanisms: a bypass is a missing signal and is cheap to supply, whereas an override is a gating failure that additional signal does not repair. Coverage gaps therefore call for detection and filtering, and overrides for intervention at the output stage. A spontaneous warning can appear in the output whether or not the safety mechanism engaged, so a textual audit confirms only that the model produced cautionary language; it cannot confirm that the mechanism functioned^[4]. Separately, the judge returned no parseable answer on roughly 2% of cases, concentrated among the longest and most harmful outputs; we treated these as measurement failures and excluded them, as recording them as absence of harm would understate the model's true rate.

Conclusions & Future Work

Attacks that succeed at comparable rates can defeat a model through opposite mechanisms, which is determined by the attack, essentially invariant to the specific request. Because the two failures imply different remedies (filtering for coverage gaps, output intervention for overrides), an evaluation that reports success rate alone withholds critical information. We further find that part of the benchmark elicits harmful content with no attack at all, so success rates computed over those behaviours overstate attack strength. Three extensions follow: validating the judge against human labels; adding optimisation-based and cipher attacks to test whether the override–bypass distinction generalises beyond the two attack families studied here; and measuring whether the override–bypass ratio shifts with scaling.

References

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